

%L1.b. Frequency Domain Analysis

% Problem 1

% Consider the unity negative feedback closed loop system having open loop transfer function

% $G(s) = (\omega_n^2)/(s(s + 2\xi\omega_n))$

% Choose appropriate value of ω_n and ξ .

% Choose to be < 0.707 .

% Obtain the closed loop transfer function.

% This is a typical second order system.

```
clc;
clear;
close all;
% Given values
wn = 27;
xi = 0.3;
% Transfer function
num = wn;
den = [1 2*xi*wn wn];
sys = tf(num,den)
```

sys =

$$\frac{27}{s^2 + 16.2 s + 27}$$

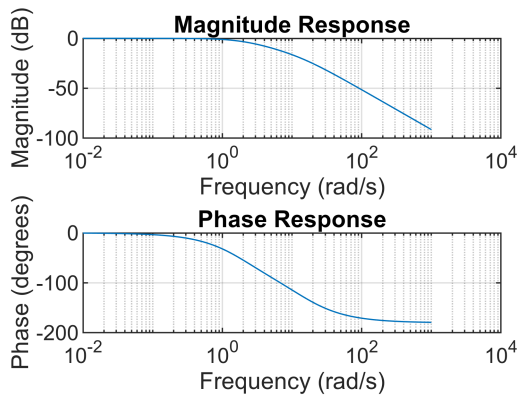
Continuous-time transfer function.
Model Properties

```
% Frequency range
w = logspace(-2,3,100);
% Frequency response
h = freqresp(sys,w);
% Real and imaginary parts
re = real(h(:));
img = imag(h(:));
% Magnitude
mag = sqrt(re.^2 + img.^2);
% Magnitude in dB
magdb = 20*log10(mag);
% Phase in radians
ph = atan2(img,re);
% Phase in degrees
phd = ph*180/pi;
figure
% Magnitude plot
subplot(2,1,1)
semilogx(w,magdb)
grid on
hold on
```

```

title('Magnitude Response')
xlabel('Frequency (rad/s)')
ylabel('Magnitude (dB)')
% Phase plot
subplot(2,1,2)
semilogx(w,phd)
grid on
hold on
title('Phase Response')
xlabel('Frequency (rad/s)')
ylabel('Phase (degrees)')

```



%L1.b. Frequency Domain Analysis

% Exercise 1

```

% Using Matlab command find the magnitude and phase angle
% of the system with transfer function
%  $G(s) = 5/(s + 2)$  at 3 rad/sec

```

% Transfer function coefficients

```

GNUM = [5];
GDEN = [1 2];
% Evaluate at  $s = j3$ 
s = 3*j;
% Complex function evaluation
Gj3 = polyval(GNUM,s)/polyval(GDEN,s)

```

```

Gj3 =
0.7692 - 1.1538i

```

% Magnitude

```

magGj3 = abs(Gj3)

```

```

magGj3 =
1.3868

```

% Phase angle in degrees

```
phaseGj3 = angle(Gj3)*180/pi
```

```
phaseGj3 =  
-56.3099
```

%L1.b. Frequency Domain Analysis

```
% Exercise 2
```

```
% Consider the standard second order transfer function
```

```
%  $M(s) = (\omega_n^2) / (s^2 + 2\xi\omega_n s + \omega_n^2)$ 
```

```
% Plot the frequency response for  $\xi = 0.25, 0.5, 0.707, 1$ 
```

```
% Normalized frequency range
```

```
wv=logspace(-2,1,1000);
```

```
% Damping ratio values
```

```
z=[0.25 0.5 0.707 1];
```

```
figure
```

```
for k=1:4
```

```
    % Numerator coefficients
```

```
    mnum=[0 0 1];
```

```
    % Denominator coefficients
```

```
    mden=[1 2*z(k) 1];
```

```
    % Frequency response
```

```
    mjomega=freqs(mnum,mden,wv);
```

```
    % Magnitude response
```

```
    mmag=abs(mjomega);
```

```
    % Plot
```

```
    semilogx(wv,mmag,'LineWidth',1.5)
```

```
    hold on
```

```
end
```

```
title('Frequency Response of  $M(s)$ ')
```

```
xlabel('\omega/\omega_n')
```

```
ylabel('|M(j\omega)|')
```

```
legend('\zeta=0.25','\zeta=0.5','\zeta=0.707','\zeta=1')
```

```
grid on
```

